

Deep learning introduction



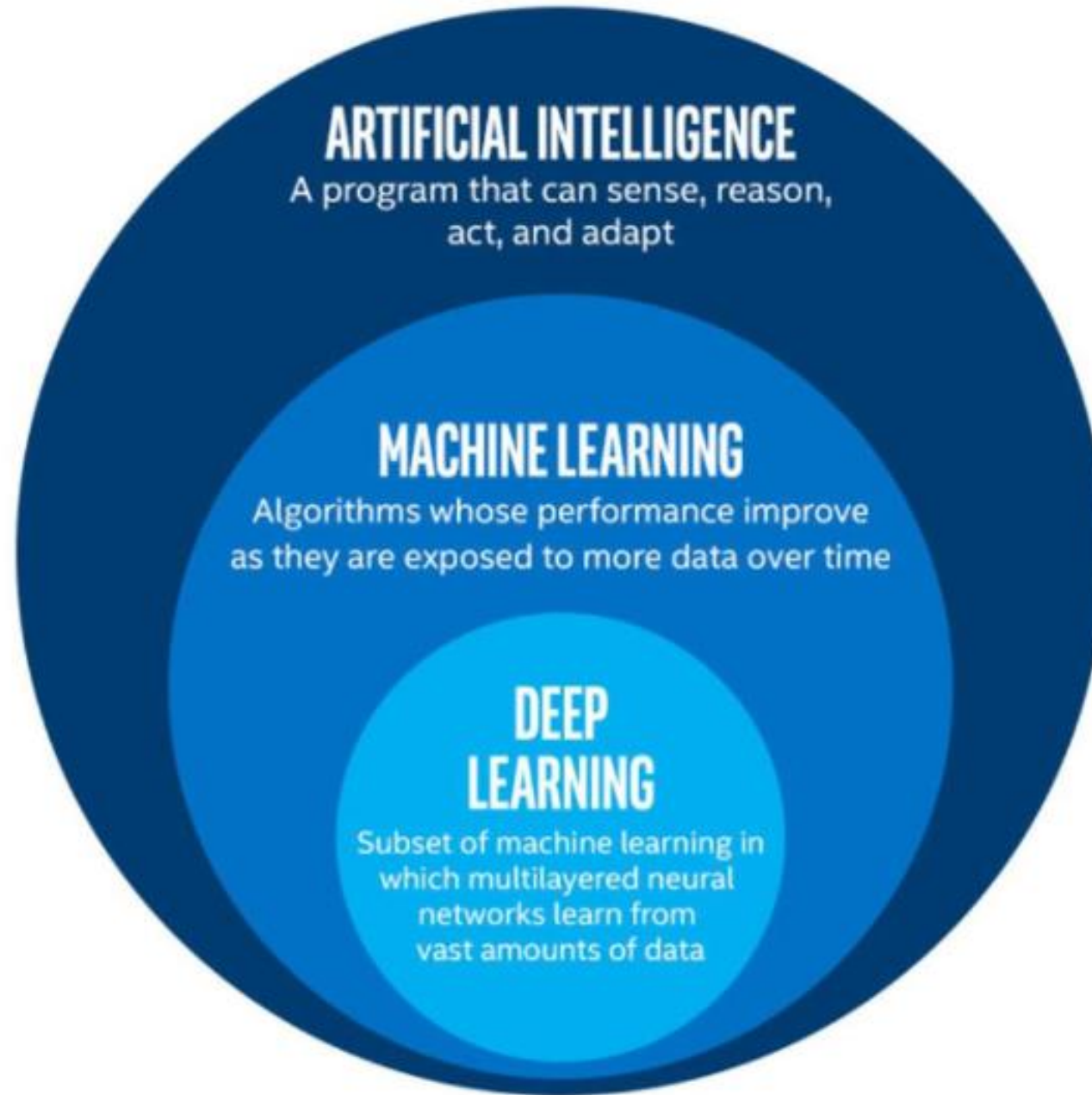
netherlands
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Rationale

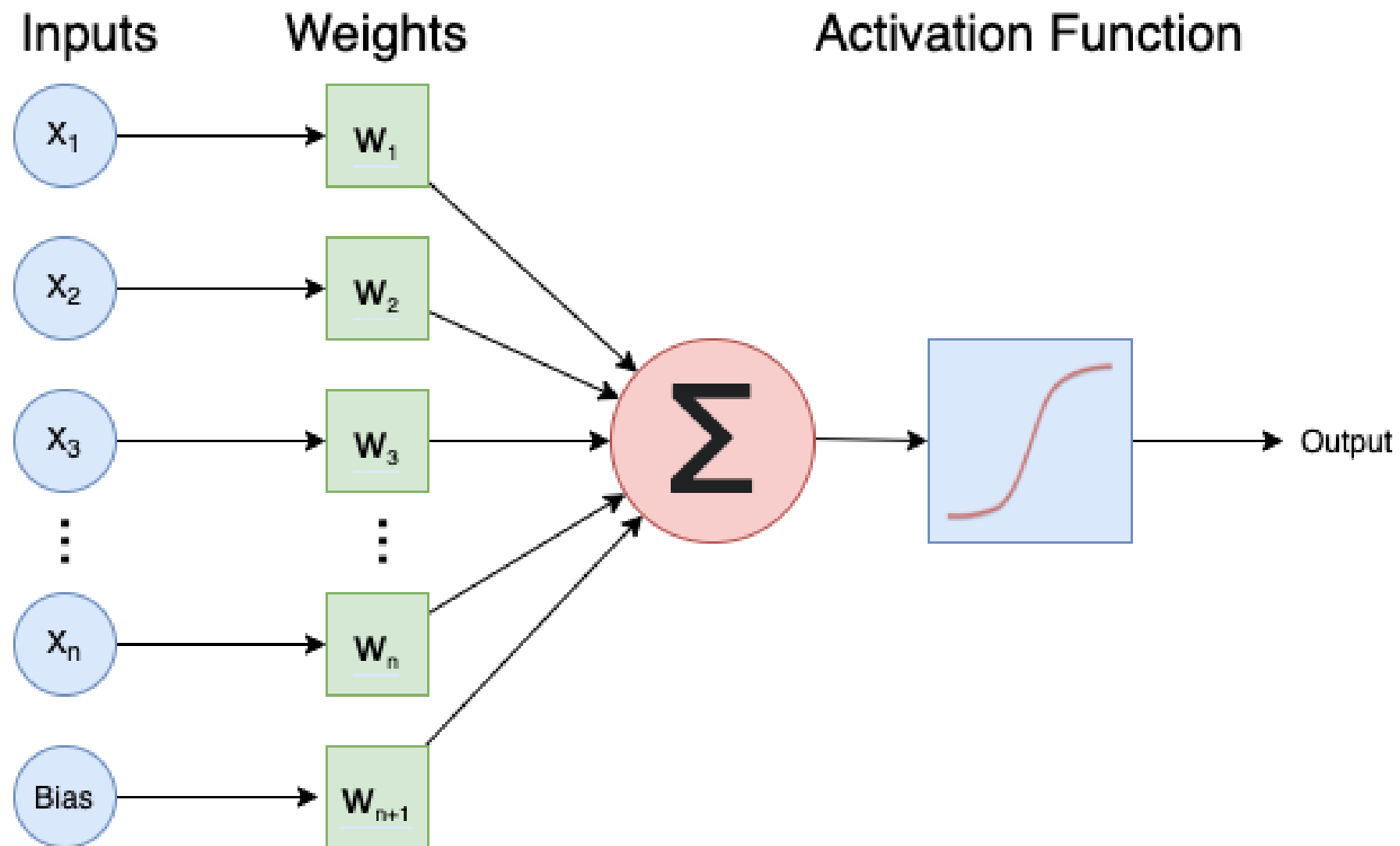
Provide you with just enough intuition about deep neural networks to start implementing them.

Then learn by doing.





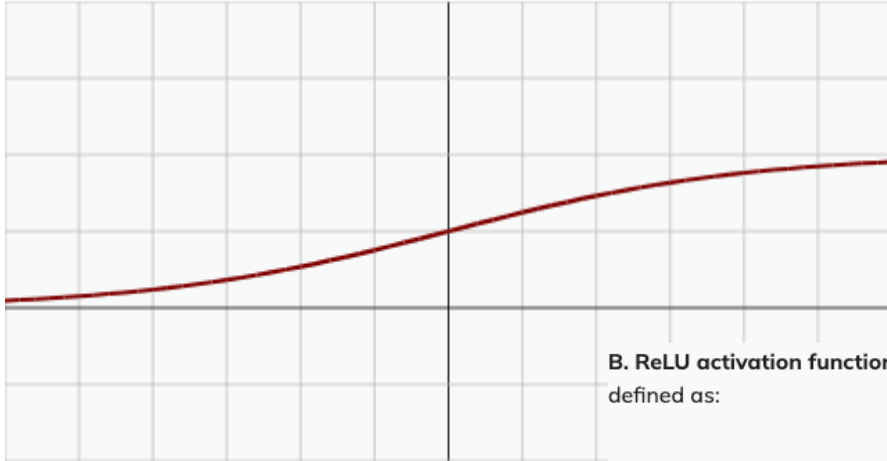
The *Neuron*



The Activation Function

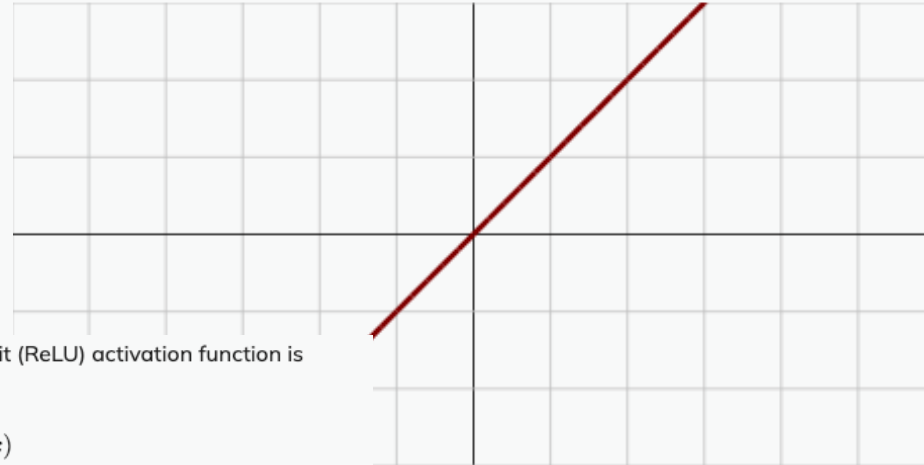
A. Sigmoid activation function The sigmoid activation function is given by:

$$f(x) = \frac{1}{1 + e^{-x}}$$



C. Linear (or identity) activation function (output=input) The linear activation function is simply the identity function:

$$f(x) = x$$



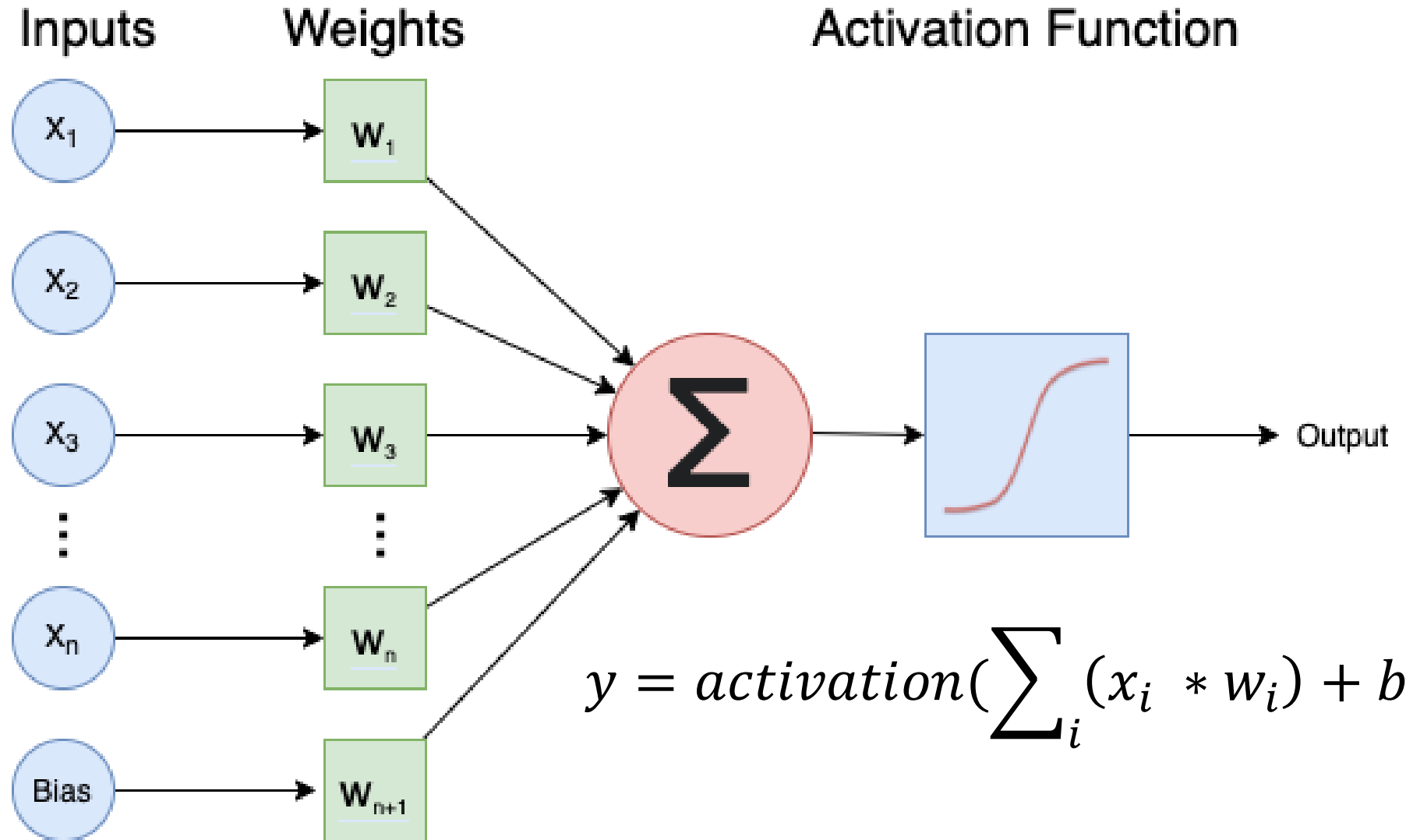
B. ReLU activation function The Rectified Linear Unit (ReLU) activation function is defined as:

$$f(x) = \max(0, x)$$

This involves a simple comparison and maximum calculation, which are basic operations that are computationally inexpensive. It is also simple to compute the gradient: 1 for positive inputs and 0 for negative inputs.



The Neuron



Calculate the output for one neuron

Suppose we have

- Input: $X = (0, 0.5, 1)$
- Weights: $W = (-1, -0.5, 0.5)$
- Bias: $b = 1$
- Activation function *relu*: $f(x) = \max(x, 0)$

What is the output of the neuron?

Note: You can use whatever you like: brain only, pen&paper, Python, Excel...

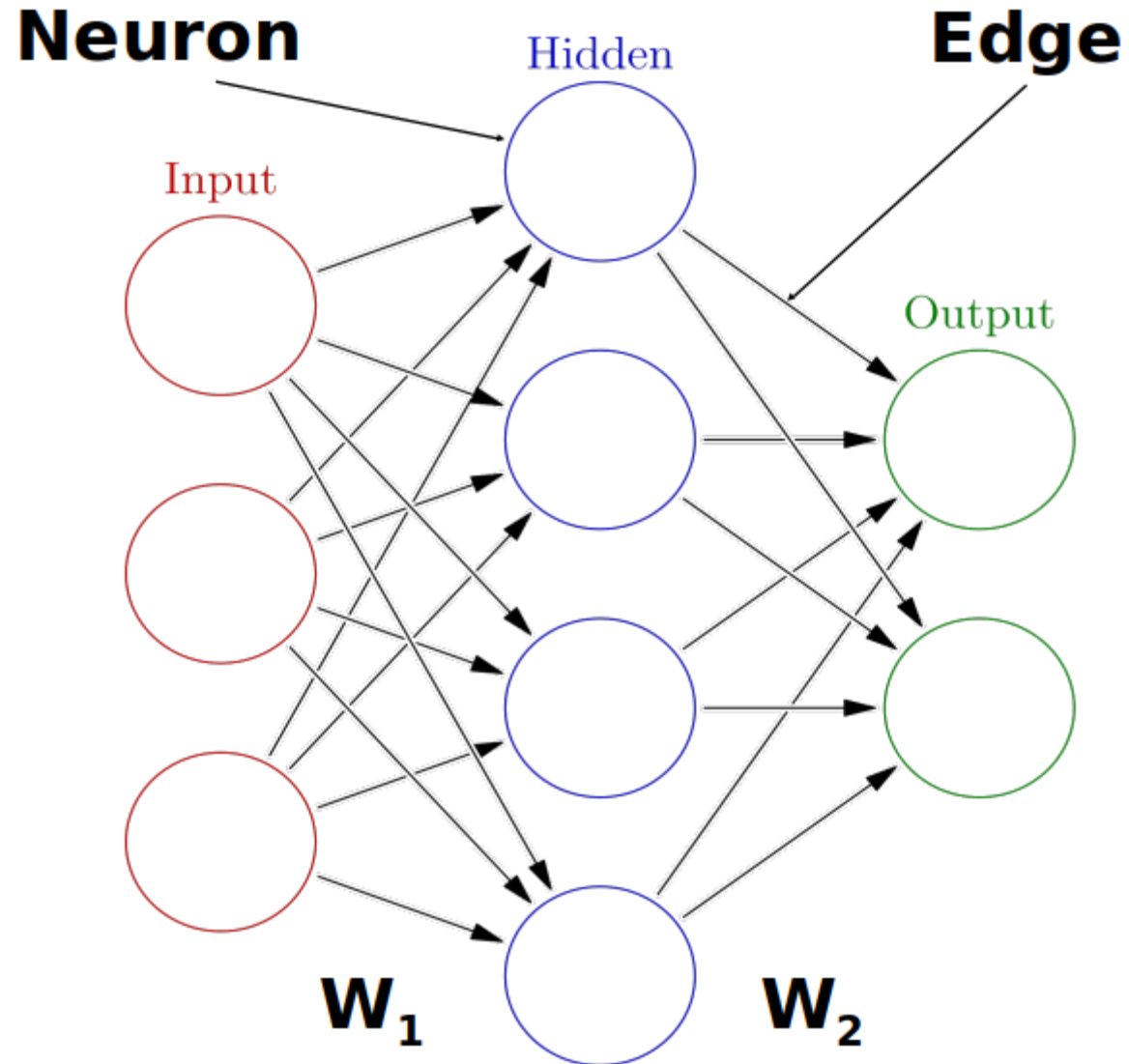


Networks as a team of detectives

- Each neuron has a certain expertise, focusing on particular patterns in the evidence (the input data)
- Together they make a decision on who is guilty.



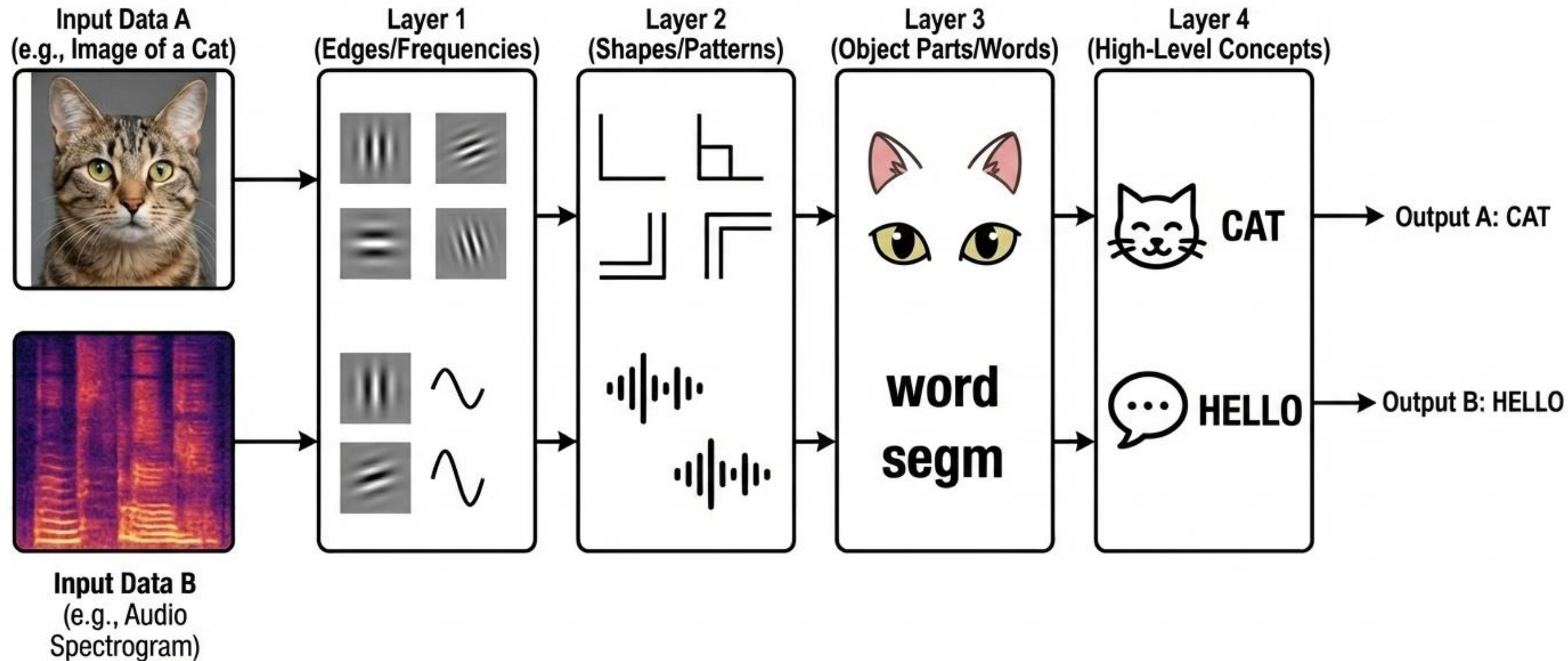
The *Neural Network*



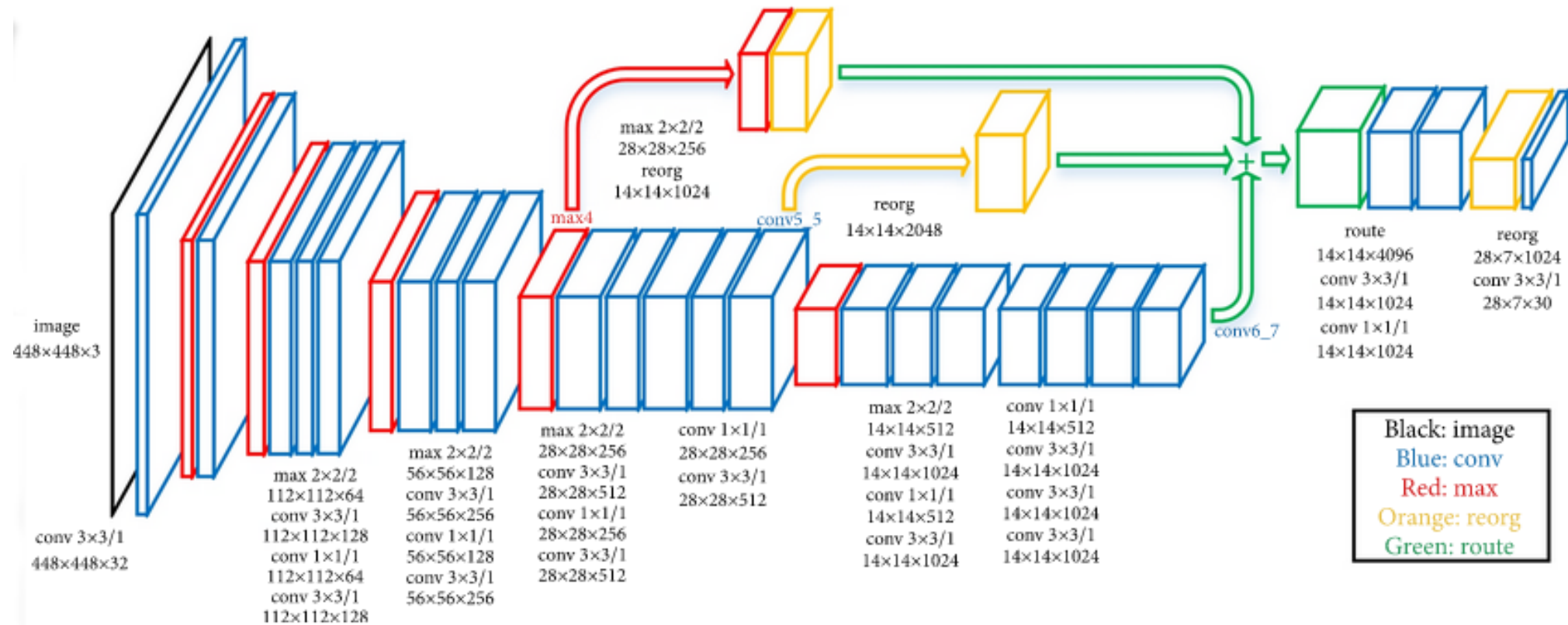
Why deep neural networks? Intuition.

- No hidden layer, simple dataset
- No hidden layer, complex dataset
- Hidden layer, complex dataset

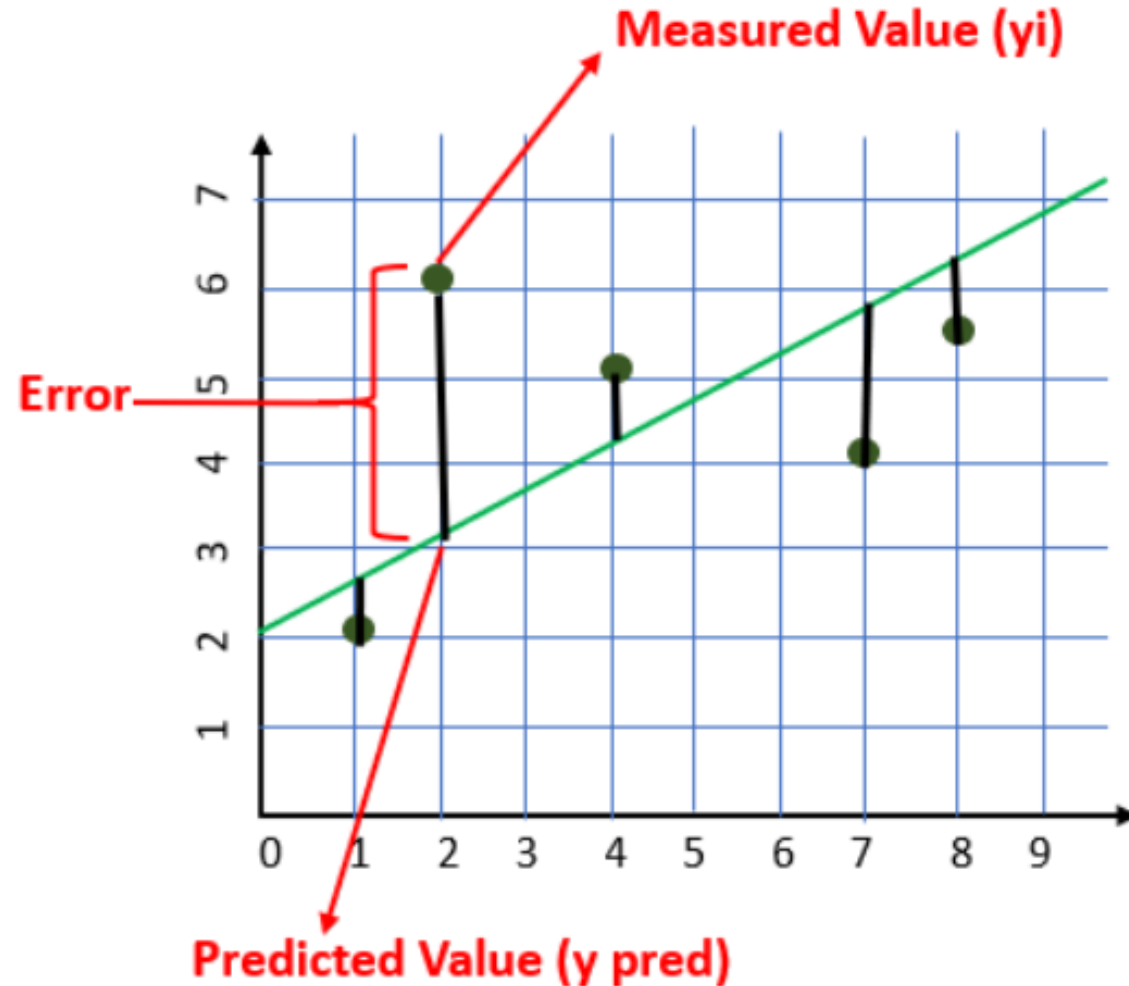




Example deep network



How do neural networks learn?



1. COMPUTE THE MEAN SQUARED ERROR

One of the simplest loss functions is the Mean Squared Error. $MSE = \frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2$. It is the mean of all squared errors, where the error is the difference between the predicted and expected value. In the following table, fill in the missing values in the 'squared error' column. What is the MSE loss for the predictions on these 4 samples?

Prediction	Expected value	Squared error
1	-1	4
2	-1	..
0	0	..
3	2	..
MSE:		..



What sort of problems can deep learning solve?

- Pattern/object recognition



DOG



CAT

What sort of problems can deep learning solve?

- Segmenting images (or any data)



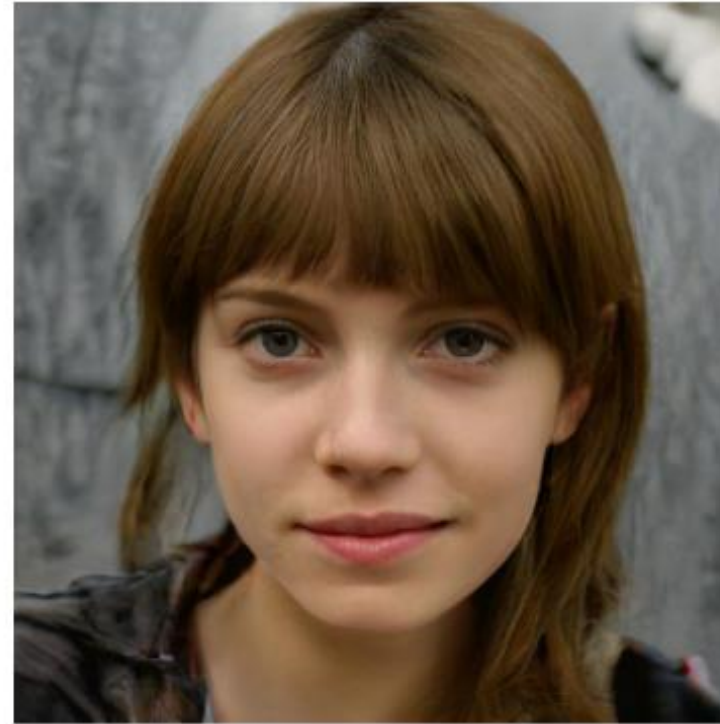
What sort of problems can deep learning solve?

- Translating between one set of data and another, for example natural language translation.



What sort of problems can deep learning solve?

- Generating new data that looks similar to the training data, often used to create synthetic datasets, art or even “deepfake” videos.



What are examples that you have of deep learning applied to research?



Problems deep learning *cannot* solve

- Any case where only a **small amount of training data** is available.
- Tasks requiring an **explanation** of how the answer was arrived at.
- Classifying things which are nothing like their training data.



Problems for which DL is inefficient

- **Logic operations**, such as computing totals, averages, ranges etc.
- Modelling **well defined systems**, where the equations governing them are known and understood.
- **Basic computer vision tasks** such as, decreasing colour depth or blurring an image.



DEEP LEARNING PROBLEMS EXERCISE

Which of the following would you apply deep learning to?

1. Recognising whether or not a picture contains a bird.
2. Calculating the median and interquartile range of a dataset.
3. Identifying MRI images of a rare disease when only one or two example images available for training.
4. Identifying people in pictures after being trained only on cats and dogs.
5. Translating English into French.



Deep learning workflow

1. Formulate/outline the problem
2. Identify inputs and outputs
3. Prepare data
4. Choose a pre-trained model or build a new architecture from scratch
5. Choose a loss function and optimizer
6. Train the model
7. Perform a Prediction
8. Measure Performance
9. Tune hyperparameters
10. Share model



Deep learning workflow exercise

Think about a problem you'd like to use Deep Learning to solve. As a group, pick 1 problem.

1. What do you want a Deep Learning system to be able to tell you?
2. What data inputs and outputs will you have?
3. Do you think you'll need to train the network or will a pre-trained network be suitable?
4. What data do you have to train with? What preparation will your data need? Consider both the data you are going to predict/classify from and the data you'll use to train the network.

Discuss your answers in your breakout group. Write them (concisely) down in the collaborative document.



Deep Learning Frameworks

- Tensorflow
 - Google
 - General Tensor Network Calculations
- PyTorch
 - Facebook
 - Very Fast
- Keras
 - Designed for Ease of Use
 - On top of Tensorflow
 - Good Documentation
- And many others (Theano, Caffe, etc.)

What we are using

- Keras – Deep Learning Framework
- Sklearn – Machine Learning Framework
- Seaborn – Visualization Library

What have we learned?

- What deep learning is
- How neural networks work
- When to apply deep learning
- Applying deep learning with the ‘deep learning workflow’



You are pioneers



Check versions

```
from tensorflow import keras  
print(keras.__version__)
```

```
import seaborn as sns  
print(sns.__version__)
```

```
import sklearn  
print(sklearn.__version__)
```

